This Zenodo deposit contains a publicly available description of the Dataset:

"Single-nucleus RNA sequencing of miBrains, a human brain tissue model".

Dataset Description:

This dataset contains snRNA-seq data generated from human induced pluripotent stem cell-derived 3D brain tissue. Specifically, this is a multicellular integrated human brain tissue (miBrain) containing multiple brain cell types. The dataset includes three experimental conditions: control miBrains, miBrains with overexpression and mutation of the SNCA gene (A53T) and the APOE3/3 genotype, and miBrains with SNCA-A53T and APOE4/4 genotype. The SNCA manipulation is present only in the neurons, whereas the APOE genotype manipulation is identical in all cell types within the same sample. Control miBrains express endogenous SNCA and harbor APOE3/3. All miBrains are isogenic.

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Dataset Name: vangheluwe-ipsc-sn-rnaseq-3dorga-apoe-asyn, v0.1

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ASAP Lab:

ASAP Project: Implications of Polyamine and Glucosylceramide Transport for Parkinson's Disease (IMPACT-PD)

Project Description: Our research program investigates through which mechanisms genetic variants associated with Parkinson's Disease modulate neurodegenerative phenotypes. As part of the ASAP Collaborative Research Network (CRN), our team generated a novel synucleinopathy-modeling platform by combining tissue engineering with genetic editing approaches. This project presents transcriptomic data of these 3D brain tissues, or miBrains, with or without synuclein pathology specifically in the neurons, and disease risk variants in the APOE gene across all cell types. By generating transcriptomic datasets of these tissues, we aim to map cell-autonomous and non-cell autonomous cellular pathways linking genetic risk variants to synuclein pathological mechanisms in a multicellular brain tissue context. Ultimately, this work aims to identify PD causal pathways and therapeutic targets to prevent neurodegeneration.

Submission Date: 2026-07-23

This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the User Manual.

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset